

# DISASTER MANAGEMENT

## CLIMATOLOGICAL DISASTERS

MODULE - 8





# DISASTER MANAGEMENT

## DISASTERS ARISING FROM NATURAL HAZARDS

### GEOPHYSICAL

- Earthquakes
- Volcano
- Tsunami



### HYDROLOGICAL

- Avalanche
- Coastal Erosion
- Coastal Flood
- Hydrological flood
- Flash Flood
- Debris/mud flow



### METEOROLOGICAL

- Cyclone
- Storm Surge
- Heat wave/ cold wave
- Heavy rains, Sand or dust storm



### CLIMATOLOGICAL

- Droughts
- Extreme hot or cold conditions
- Forest fires
- Glacial lake outburst



### BIOLOGICAL

- Epidemics – viral, bacterial, parasitic, fungal or prion infections
- Insect infestations
- Animal stampedes





## COASTAL EROSION IN INDIAN BEACHES



- 1** Indian Coasts serve multidimensional needs such as recreational purpose, seaport for maritime commerce, residential and commercial structures, etc.
- 2** With the growth of population in the coastal zone, more competing pressures are being put on the shores and beaches.
- 3** Developmental activities along India's coastal areas have increased extensively in recent decades and this trend is expected to continue. Due to this the erosion rate has increased.



## COASTAL EROSION - CAUSES AND IMPACTS

**1** Abrasive action along sand and gravel moved by the waves.

**2** Destruction of mangroves, which act as a natural safeguard.

**3** Flooding of river during monsoon also lead to inundation and causes sea erosion.

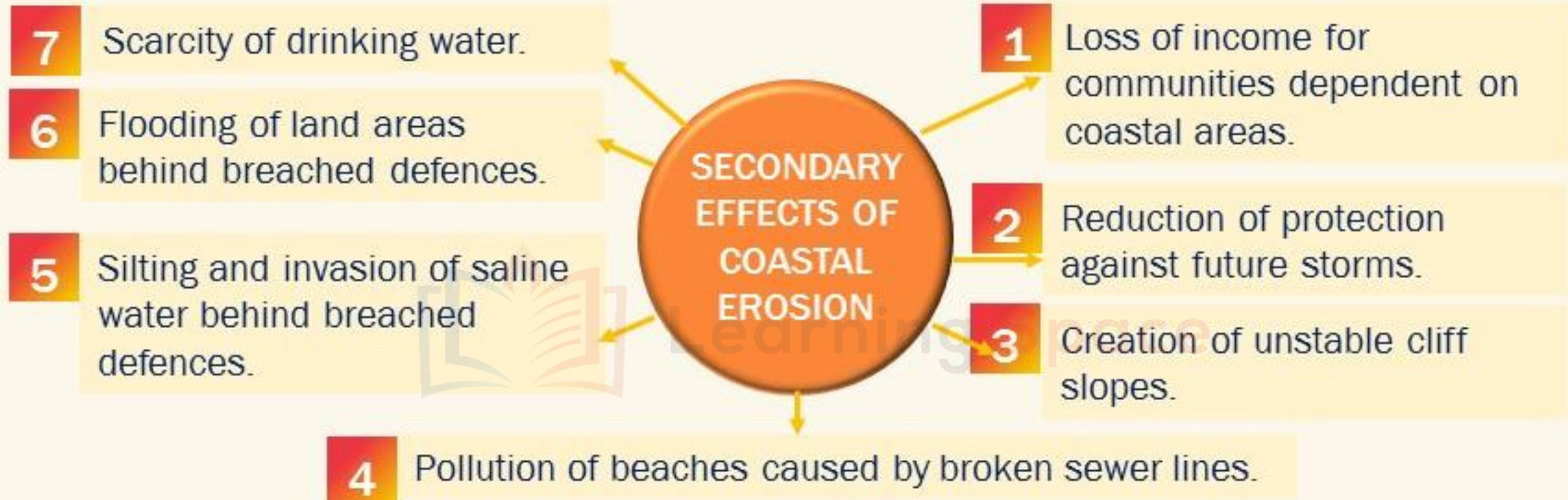
**1** Weakening and destruction of sea walls, revetments, bulkheads and other coastal defenses.

**2** Loss of agricultural land and destruction of existing habitats of shore flora and fauna.

**3** Loss of Beach sediments and Deposition of sediment on productive lands









## WHAT ARE THE WAYS TO MITIGATE COASTAL EROSION

- 1 Offshore breakwaters to reduce the face of tidal surges before these reach the land
- 2 Seawalls to reduce flooding inland; including the initial surges
- 3 Construct concrete floodwalls to direct floodwaters away from populated areas
- 4 Bulkheads, to hold landfill in place.
- 5 Revetments, usually, large- stones, on relatively flat slopes to slowdown flooding.
- 6 The "Soft" Engineering Approach includes: Sand fills; Sand fences; Beach grasses and Salt marshes.
- 7 In all vulnerable coastal areas the Coastal Zone Management should be effectively introduced.
- 8 The traditional knowledge of the local people should be tapped and used for sustainable development.
- 9 Afforestation programme with suitable species.
- 10 The "passive" or non-structural approaches include using zoning regulations to limit -development.
- 11 Using building codes to reduce damage from flooding and winds





# DISASTER MANAGEMENT

## COASTAL FLOOD RISK



- 1 A coastal flood starts when waves move inland on an undefended coast or overtop or breach the coastal defence works like dunes and dikes.
- 2 As climate change will put three times more people at risk of coastal flooding by 2050 than previously thought
- 3 Mumbai and Kolkata are on the list of global cities that face the risk of being wiped out, according to a latest research report
- 4 A recent IPCC report had warned that a mere 50 cm rise in sea level is enough to flood major port cities around the world.





# WHAT ARE THE MAJOR REASONS FOR COASTAL FLOOD?

## RECLAIMING THE LAND

- This is land that has been gained from the sea due to coastal management. This land is low lying and flat, so a small rise in sea level from a mild storm surge is enough to flood it and cause extensive damage.
- For example, reclaimed land is occupied by the majority of Mumbai population making it vulnerable to coastal flooding.



## RAPID URBANISATION

- The inadequate drainage system and an ineffective spatial planning increase the flood risk.
- This also increases with sea level rise, geology, and rapid coastal population growth accompanied by rapid increase of human activities that interfere with natural processes.





# WHAT ARE THE MAJOR REASONS FOR COASTAL FLOOD?

## INCREASE INFRASTRUCTURE IN COASTAL ZONES

Construction activities in ecologically sensitive zones and near the 'high tide line' has increased the vulnerability.



## MANGROVE DEGRADATION

Mangroves serve as a natural barrier to Tsunamis and cyclones. Cutting of mangroves to make way for building infrastructure has curtailed this natural resilience.





## WAYS TO MITIGATE COASTAL FLOOD

- 1 Shelter buildings, especially designed for cyclone conditions should be constructed in the coastal areas.
- 2 Hazard zoning and mapping, including the use of GIS tools, capacity development, etc; and its implementation strategies are suggested.
- 3 The National Centre for Sustainable Coastal Management in demarcating the High Tide Line and protect ecologically sensitive areas from development
- 4 Hedges and other plantations, wherever feasible, should be grown as wind breakers for a width of about 2 km along the seacoast
- 5 Only restricted urbanisation should be permitted in areas vulnerable to sea surges resulting from cyclones
- 6 The general range of land use regulation recommended for flood prone areas and to regulate the menace of sand mining activities.





## EARLY WARNING SYSTEMS FOR COASTAL FLOOD AND SOME RELATED CONCERNS

- 1** National Institute of Ocean Technology under the Ministry of Earth Sciences, is engaged in data collection for tsunami warning and beach restoration which is of immense benefit to the society
- 2** Coastal Flood Warning System app for Chennai (CFLOWS- CHENNAI) developed by the National Institute of Ocean Technology.
- 3** The Disaster Management Support (DMS) Programme of ISRO: As part of R&D support to DMS for remote sensing applications.
- 4** The latest notification drastically shrinks 'No Development Zone (NDZ)' to 50 m from HDL in densely populated areas (where population exceeds 2,161 per sq km as per the 2011 Census).
- 5** No study is available to show the carrying capacity of coastal areas to accommodate such increased development.
- 6** The government has declared Sagarmala, Bharatmala and CEZs as "strategic projects", which has a blanket exemption from CRZ provisions.





## DROUGHT RISK



- 1** Drought is generally considered as a deficiency in rainfall /precipitation over an extended period.

- 2** Droughts results in a water shortage causing adverse impacts on vegetation, animals, and/or people.

- 3** Time-lines have been indicated for the declaration of drought, namely, 30 October for Kharif and 31 March for Rabi.

- 4** India Meteorological Department (IMD) has replaced the word “drought” to describe poor rainfall with “deficient year” and “large deficient year”.





# DISASTER MANAGEMENT

## DROUGHT RISK

### METEOROLOGICAL DROUGHT

The actual rainfall in an a smaller area is significantly less than the climatological mean of a larger area.



### HYDROLOGICAL DROUGHT

A marked depletion of surface water causing very low stream flow and drying of lakes, rivers and reservoirs



### AGRICULTURAL DROUGHT

Inadequate soil moisture resulting in acute crop stress and fall in agricultural productivity







**INDIA**  
DROUGHT PRONE AREAS

 Drought Prone Area



## THE DROUGHT HAZARD...A TIMELINE

### PRE-INDEPENDENCE

- 1 Droughts during the colonial period tended to degenerate into severe famines causing massive human losses.
- 2 In the latter half of the 19<sup>th</sup> century, there were approximately 25 major famines across India, which killed 30-40 million people.

### POST - INDEPENDENCE

- 1 The situation improved remarkably in post-independent India DUE TO Investment in irrigation works, promotion and availability of quality inputs.
- 2 Focus on research & extension led to increased agricultural productivity and greater resilience among the farming communities.

### POST - LIBERALIZATION

- 1 With the liberalization of the Indian economy in the 1990s, accelerated growth in industry and services saw the share of agriculture in GDP shrink to less than 15%
- 2 Yet the country continued to be largely self-sufficient in food and agri-commodities, gained greater resiliences in absorbing the impact of drought.





# REDUCING DROUGHT IMPACTS AND BUILDING RESILIENCE





# PREPAREDNESS OF DROUGHT HAZARDS THROUGH GOVERNMENT PROGRAMMES



**1** Seeing watershed as key biophysical unit for drought proofing & resilience.

**2** The Green India Mission – aims to restore 10 M ha in 10 Years. Under the Bonn Challenge India's pledge to restore its 13 million hectares of degraded land by 2020.

**3** The focus on restoration of ecosystem services, using landscapes and watershed units.

**4** Pilot project to restore its degraded forests in five states Haryana, Madhya Pradesh, Maharashtra, Nagaland and Karnataka.

**5** Integrated Watershed Management Programme – targeted development of 75 million hectares of rainfed/degraded area in a phased manner during 2007-2027.

**6** The Mahatma Gandhi National Rural Employment Guarantee Scheme – strong focus on land, water and afforestation activities

**7** The mission for Sustainable Agriculture – including climate-smart agriculture, National Food Security Mission, National Mission on Micro Irrigation among others





## GLACIAL LAKE OUTBURST FLOOD (GLOF)



**1** The glacier floods caused by the drainage of naturally dammed lakes in the glacier, on or at the margin of glaciers.

**2** Retreat of glaciers in the wake of global warming, mass tourism; developmental interventions, Black carbon are increasing the risk of GLOF.

**3** The Hindu Kush-Himalayan region has suffered several GLOF events originating from numerous glacial lakes, some of which have trans-boundary impacts.

**4** But on the other hand, glacial lakes may offer considerable benefits to local community. They can provide a natural storage facility for water as water supplies becoming increasingly scarce





## STEPS TAKEN TO CONTROL GLACIAL OUTBURST FLOODS



**1** Glacial Lakes Outburst Floods (GLOFs), are a subject of concern in the Sikkim Himalayan region as several lakes have been formed due to melting of scores of glaciers in the region.

**2** Sikkim has installed a Lake monitoring and information System (water level Sensor) at South Lhonak lake. The sensor gives the water level of the lake and also monitored the lake level when there is sudden fluctuation in water level

**3** To improve food security and reduce flood risks due to deforestation and inefficient water, women should be trained in home gardening, water-efficient farming technologies should be installed along with large scale reforestation programmes.





# DISASTER MANAGEMENT

## FOREST FIRES AROUND THE WORLD





# FOREST FIRES ARE INCREASING IN THE WORLD !



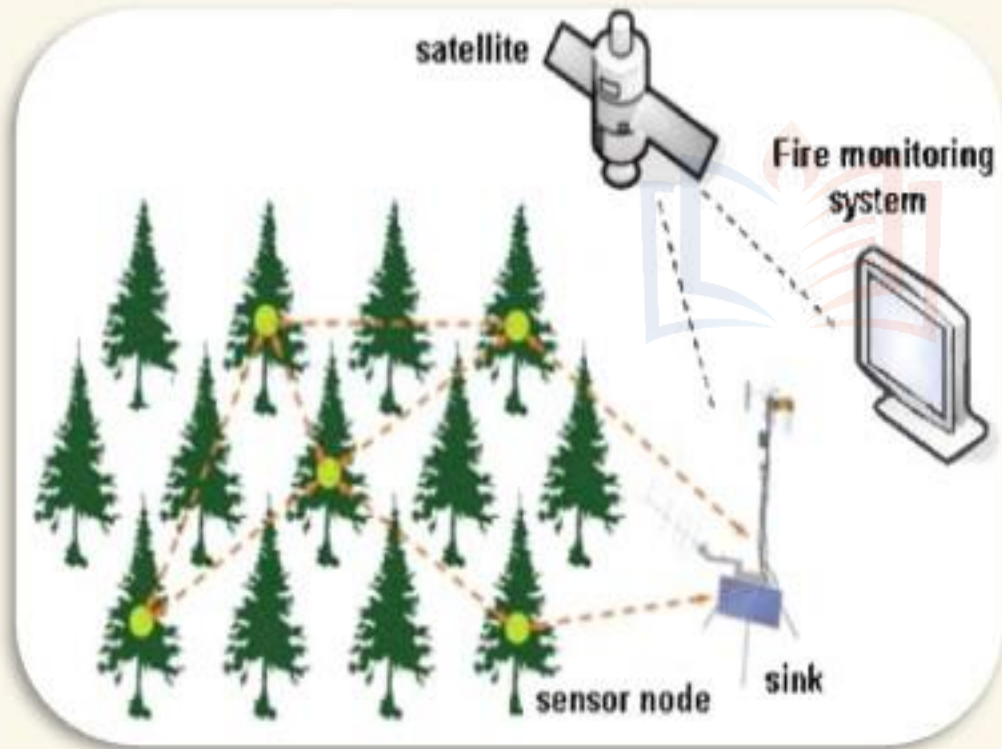
**1** Fires are very important landscape management tools and are used to clear land for planting crops which can get aggravated by higher than normal temperatures

- 2** The main reason for the spike in these large fire incidences is increase in temperature and rain fall deficit.
- 3** Vast tracts of Amazon forest in Brazil, Russia , Siberia (being the coldest region) Africa have also been burning. The extent of the fires has sparked outrage around the world.
- 4** Experts believe the majority of fires across Brazil this year are caused by human activity - farmers and loggers clearing land for crops or grazing, as 2019 is not as dry compared with other years.





## EFFORTS TO MITIGATE FOREST FIRES



**1** Development of systems for early warning and fire danger rating.

**2** Digitization of management boundaries by the state forest departments.

**3** Improvising giant water balloons that are GPS aided or managed by any app to extinguish the forest fires as early as possible.

**4** Sensitization of communities should be done to ensure that fire is used responsibly

**5** Provision of training should extend beyond state-managed forests to community institutions





## EFFORTS TO MITIGATE FOREST FIRES



- 6** The Sentinel-3 World Fires Atlas Prototype product has been developed by European Space Agency to identify all active fires at night.
- 7** The report, 'Strengthening Forest Fire Management in India', jointly prepared by the Ministry of Environment, Forest and Climate Change (MoEFCC) and the World Bank for tackling and understanding the menace of forest fires.
- 8** To mitigate such events, National Mission for Green India under India's National Action Plan on Climate Change, the government has committed to increase forest and tree cover.
- 9** Since forest fires are not typically treated as disasters, the disaster management authorities have so far played a minor role in forest fires.
- 10** Stronger collaboration between the State Forest Departments (SFDs), the disaster management authorities and research entities are required.







## RELATED QUESTIONS THAT CAN BE ANSWERED AFTER READING THE MODULE



- 1** Examine the events of sinking islands and coastal erosion in recent times. Give necessary measures to overcome the situations.
- 2** Elaborate the forest fires in Amazon rainforests. What steps can be taken to control the forest fires.
- 3** Analyse the drought map of India and give region specific solutions. How the national framework is prepared to mitigate the effects of drought?
- 4** Discuss the causes and effects of the Glacial Lakes Outburst floods or GLOFs. Why are they increasing in threat for coming years. What are the steps needed to mitigate the threats?





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